Understanding .lib Files

INTRODUCTION:

This guide provides an introduction to reading and understanding a Liberty file (.lib), which is a standard format used in electronic design automation (EDA) to describe the characteristics of a cell library. This file contains critical information about the timing, power, and physical properties of the logic gates and other components used in a digital circuit design.

OBJECTIVES:

After completing this lab, you will be able to:

- Identify the key sections within a Liberty file.
- Understand the purpose of different attributes and their values.
- Interpret timing and power data for a given cell.
- Locate information about pin characteristics and logical functions.

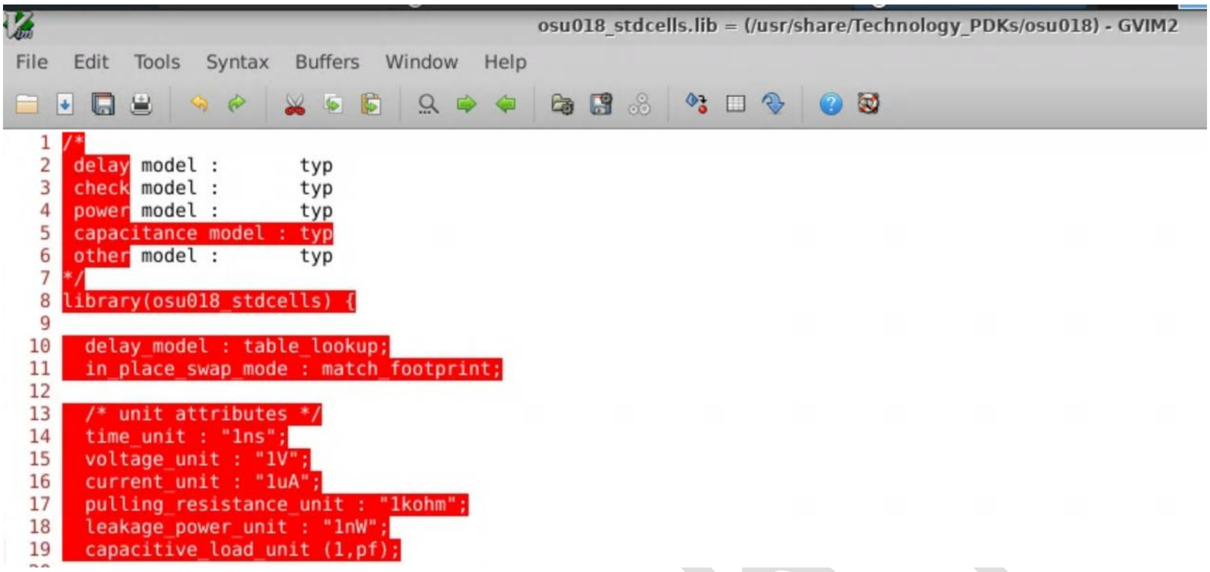
PROCEDURE:

A Liberty file is typically organized in a hierarchical structure, with different sections and attributes defining the library's properties and the individual cells within it.

This lab consists of the following steps:

Section 1: Library-Level Attributes The file begins with attributes that apply to the entire library. These may include:

- library(): The main block that encapsulates all library information.
- technology: Specifies the semiconductor technology, e.g., "cmos."
- date: The date the library was generated.
- revision: The revision number of the library.
- delay_model: The timing model used, e.g., "nldm" (Non-Linear Delay Model).
- time_unit, voltage_unit, current_unit, capacitive_unit: Defines the units for various measurements within the file.



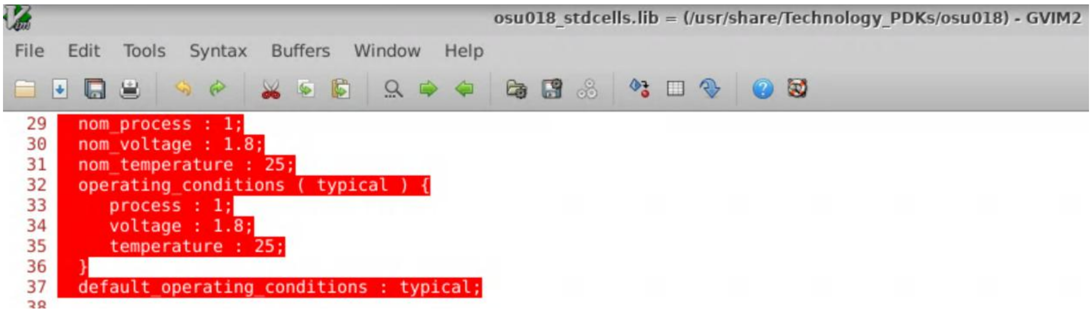
Section 2: Operating Conditions and Wire Load Models This section describes the conditions under which the library was characterized and the models used for interconnect parasitics.

- operating_conditions(): Defines parameters like process (process), voltage (voltage), and temperature (temperature).
- wire_load_model(): Provides a model to estimate the resistance and capacitance of interconnects based on fanout

  Example:

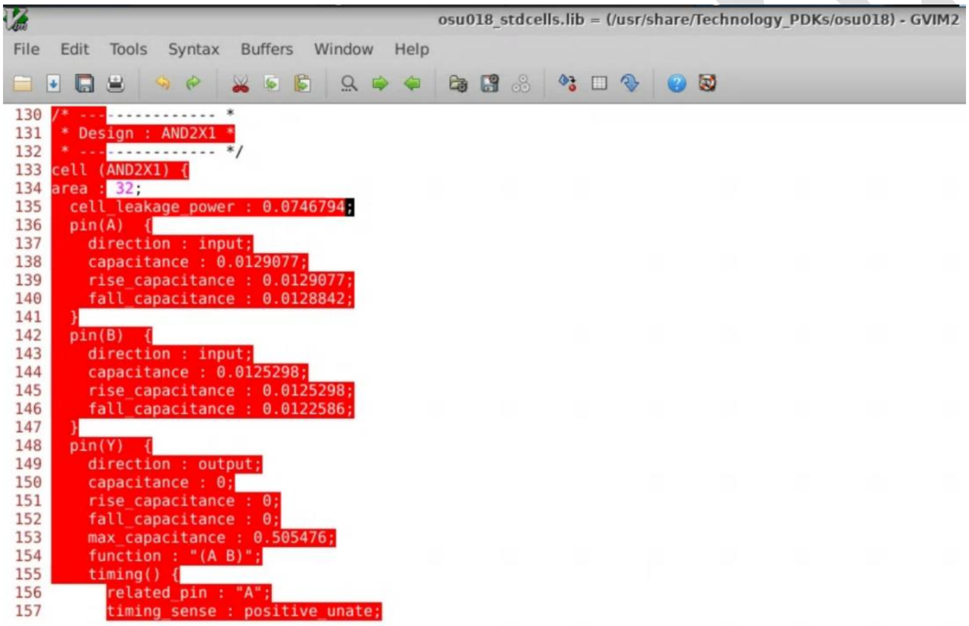
```
operating_conditions("typical_conditions") {
  process : 1.0;
  voltage : 1.2;
  temperature : 25;
}

wire_load_model ("WLM_10X") {
  resistance  : 0.1;
  capacitance : 0.05;
  fanout_length (1, 10);
  fanout_length (2, 20);
}
```



Section 3: Cell Definitions The cell() block is the most detailed section, containing all the information for a specific logic gate or component.

- cell(cell_name): The main block for a single cell, e.g., cell(AND2X1).
- area: The physical area of the cell.
- ff or latch or nand, etc.: The functional definition of the cell.



EXPLANATION:

- cell(AND2X1): Defines a cell named "AND2X1", which is a 2-input AND gate.
- area: The physical area of the cell is 10 (in whatever units were defined).
- pin(A) and pin(B): These are the two input pins. Each has an input capacitance of 1.
- pin(Z): This is the output pin. Its function is defined as "A & B".

CONCLUSION:

In this guide, you learned how to read and understand a Liberty file. By breaking down the file into its main sections and exploring the purpose of lookup tables, you can interpret the timing and power data for digital components. This knowledge is essential for working with EDA tools in circuit design.